

SOUTH WEYMOUTH NAS, MA, USA

WMO: 725097

Lat: 42.150N

Lon: 70.933W

Elev: 49

StdP: 100.74

Time Zone: -5.00 (NAE)

Period: 82-93

WBAN: 14790

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.5	(c) -12.0	(d) -25.5	(e) 0.4	(f) -12.4	(g) -22.3	(h) 0.5	(i) -10.0	(j) 8.9	(k) -0.3	(l) 8.2	(m) -1.9	(n) 2.9	(o) 300	(p) 0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.9	23.2	31.0	22.4	29.3	21.5	25.0	30.5	23.8	28.8	22.8	27.4	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 23.4	(b) 18.3	(c) 27.7	(d) 22.3	(e) 17.1	(f) 26.3	(g) 21.3	(h) 16.0	(i) 25.3	(j) 76.1	(k) 30.7	(l) 71.5	(m) 29.0	(n) 67.5	(o) 27.4	(p) 28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.4	6.5	-18.9	35.3	2.0	1.4	-20.4	36.3	-21.6	37.1	-22.7	37.8	-24.2	38.8
			-19.7	26.6	1.9	0.9	-21.1	27.3	-22.3	27.9	-23.4	28.4	-24.8	29.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-2.1	-0.6	3.4	8.6	14.5	19.2	22.4	21.6	17.3	12.1	6.8	1.5
	DBStd	9.43	5.32	4.93	4.84	4.14	4.45	3.82	3.20	3.50	4.01	4.26	4.54	5.25
	HDD10.0	1384	376	297	213	74	10	0	0	0	1	27	119	267
	HDD18.3	3240	634	529	464	292	137	34	6	10	66	199	347	521
	CDD10.0	1547	1	1	7	33	148	275	384	360	220	91	22	4
	CDD18.3	359	0	0	0	1	17	59	131	112	35	4	0	0
	CDH23.3	3186	0	0	2	32	221	607	1137	904	260	22	2	0
	CDH26.7	1108	0	0	0	11	77	219	416	308	76	1	0	0
Wind	WSAvg	2.8	2.9	3.2	3.3	3.2	2.9	2.6	2.3	2.3	2.2	2.6	2.9	2.9
Precipitation	PrecAvg	1112	101	100	96	90	81	74	64	86	89	95	108	108
	PrecMax	1511	272	192	250	193	170	263	124	202	212	227	245	242
	PrecMin	644	12	16	18	29	13	16	17	16	12	31	19	27
	PrecStd	195	55	41	52	37	41	51	30	51	55	41	57	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.0	22.0	27.5	32.6	35.1	35.4	34.4	31.6	25.9	22.6	16.9
		MCWB	11.0	10.9	12.9	16.8	20.6	23.1	24.6	26.0	24.3	18.9	16.3	14.4
	2%	DB	10.4	12.1	16.4	21.9	28.7	31.7	32.8	32.0	29.0	23.2	19.0	14.2
		MCWB	8.2	9.1	11.2	14.3	19.4	21.8	23.6	23.6	22.2	18.1	15.0	12.1
	5%	DB	7.7	9.5	13.3	18.7	25.5	29.0	30.7	29.9	26.3	21.2	16.5	12.2
		MCWB	5.2	7.0	9.3	12.1	17.9	20.3	22.2	22.6	20.2	16.9	13.3	10.6
	10%	DB	5.6	7.2	10.8	15.6	22.6	26.9	29.0	28.1	24.0	19.3	14.5	9.5
		MCWB	3.4	4.5	7.3	10.6	15.7	19.1	21.6	21.5	19.1	15.3	11.9	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.4	14.6	17.9	23.1	24.3	26.2	26.8	25.0	21.3	18.2	15.2
		MCDB	13.7	14.1	18.7	26.1	31.2	30.9	32.0	33.4	30.7	23.0	20.3	16.8
	2%	WB	8.4	9.5	12.0	15.4	20.5	23.0	24.9	25.1	22.9	19.5	15.7	12.7
		MCDB	10.4	11.6	15.9	20.2	26.3	30.4	30.8	30.1	27.2	22.1	18.2	13.9
	5%	WB	5.7	7.5	10.0	13.4	18.6	21.6	23.7	23.9	21.7	17.8	13.8	10.5
		MCDB	7.4	9.8	13.0	17.2	23.9	27.8	28.8	27.9	25.2	20.4	16.0	12.2
	10%	WB	3.5	4.7	7.5	11.4	16.8	20.2	22.7	22.8	20.0	16.1	12.0	7.6
		MCDB	5.3	6.9	10.6	15.0	21.4	25.2	26.9	26.7	22.9	18.6	14.2	9.3
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.6	9.8	10.0	11.4	11.6	11.0	10.3	10.9	10.7	9.5	9.0
		MCDBR	11.1	11.9	14.3	15.4	16.1	15.3	13.6	12.5	12.5	12.2	13.1	11.1
		MCWBR	10.1	10.0	10.4	9.2	9.1	8.2	6.8	5.9	7.2	8.1	10.6	11.0
	5% WB	MCDBR	10.9	11.6	12.9	13.7	14.0	13.5	11.8	10.6	10.9	10.0	11.1	11.2
		MCWBR	10.6	10.5	10.5	9.5	8.9	8.2	6.7	5.7	7.5	7.4	10.1	11.3
Clear-Sky Solar Irradiance	taub		0.295	0.303	0.326	0.368	0.405	0.452	0.473	0.432	0.373	0.335	0.310	0.296
	taud		2.510	2.498	2.487	2.394	2.331	2.233	2.182	2.283	2.429	2.542	2.573	2.569
	Ebn at Noon		861	909	924	901	872	828	805	828	860	860	837	831
	Edh at Noon		71	85	96	114	125	138	144	126	101	81	67	63
All-Sky Solar Radiation	RadAvg		1.77	2.65	3.65	4.71	5.25	5.67	5.94	5.34	4.30	2.88	1.95	1.47
	RadStd		0.12	0.24	0.26	0.51	0.56	0.54	0.35	0.25	0.29	0.25	0.15	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+43

Nomenclature: See separate page